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Vertical distribution and risk assessment of pharmaceuticals and other micropollutants in southern North Sea coastal waters

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ABSTRACT

Pharmaceutical compounds are micropollutants of emerging concern, as well as other classes of chemicals such as UV filters and artificial sweeteners. They enter marine environments via wastewater treatment plants, aquaculture runoff, hospital effluents, and shipping activities. While many studies have investigated the presence and distribution of these pollutants in numerous coastal areas, our study is the first to focus on their occurrence, spatial distribution, and vertical distribution in the sea surface microlayer (SML) and the near-surface layer of marine environments. We analyzed 62 pharmaceutical compounds, one UV filter, and six artificial sweeteners from the SML to the corresponding underlying water (0 cm, 20 cm, 50 cm, 100 cm, and 150 cm) at four stations in the southern North Sea. One station is the enclosed Jade Bay, one is the Weser estuary at Bremerhaven, and the other two stations (NS_7 and NS_8) are in the open German Bight. Jade Bay receives pollutants from surrounding wastewater treatment plants, while the Weser estuary receives pollutants from cities like Bremerhaven, which has dense populations and industrial activities. Concentrations of pharmaceutical compounds were higher in the upper water layers (from the SML to 20 cm). Eleven pharmaceutical compounds (caffeine, carbamazepine, gemfibrozil, ibuprofen, metoprolol, salicylic acid, clarithromycin, novobiocin, clindamycin, trimethoprim, and tylosin) were detected in >95 % of our samples. One UV filter (benzophenone-4) was found in 83 % and three artificial sweeteners (acesulfame, saccharin, and sucralose) in 100 % of all our samples. All artificial sweeteners posed high risks to the freshwater invertebrate *Daphnia magna*. Understanding the spatial and vertical distribution of pharmaceuticals and other micropollutants in marine environments may be essential in assessing their dispersal and detection in other aquatic environments.

1. Introduction

The occurrence of pollutants in marine environments is a public health concern due to their adverse effects on ecological resilience and human and animal health. The World Health Organization (2022) has developed the "One Health" framework to reduce the discharge of pharmaceutical compounds in marine environments and to prevent the spread of resistant microbes and the accumulation of micropollutants in humans, animals, and the environment. The distribution of pharmaceutical compounds and other so-called 'contaminants of emerging concern' (e.g., UV filters and food additives) in marine environments has received significant attention due to their increased utilization and continuous discharge into aquatic domains. Pharmaceutical compounds are residues of pharmaceutical drugs or their metabolites that find their way into marine environments via wastewater treatment plants (WWTP), aquaculture runoff, hospital effluents, shipping activities, and sediment remobilization (Alygizakis et al., 2016; Szekeres et al., 2017; Biel-Maeso et al., 2018; Afsa et al., 2020; Su et al., 2020; Biel-Maeso and Lara-Martín, 2021).

Pharmaceutical compounds are neither completely metabolized in

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patients before they are released into municipal WWTP influents nor do WWTP eliminate them before they are released into marine environments (Alygizakis et al., 2016). According to Alexy et al. (2006), >50 % of antibiotics (cephalosporin, carbapenem, aminoglycosides, glycopeptides, and fluoroquinolone) consumed by a local population in Kenzening (Germany) were not metabolized and were released into the municipal WWTP. Global fish production has also fostered the use of pharmaceutical compounds in fish health protection and growth maximization (Adenaya et al., 2023). According to Singh and Singh (2018), fish farmers use pharmaceutical compounds such as antibiotics and anesthetics to stimulate fish health and reduce fish stress. However, 75 % of these compounds are not metabolized in fish, resulting in their discharge into marine environments (Burridge et al., 2010; Romero et al., 2012).

The distribution of pharmaceutical compounds and other organic micropollutants largely depends on locations because different anthropogenic activities and pollutant inputs affect different marine environments (Hawash et al., 2023). Lara-Martín et al. (2014) and Alygizakis et al. (2016) already showed that the concentrations of some pharmaceutical compounds in different marine environments differed with locations and depths. However, Lara-Martín et al. (2014) sampled surface seawaters and sediments of one coastal environment. Alygizakis et al. (2016) did not sample systematically, making it difficult to conclude the vertical distribution of pharmaceutical compounds in marine environments.

Despite these baseline studies, data from the sea surface microlayer (SML), the marine environment's uppermost 1–1000 μm layer, does not exist. The SML connects marine environments and the overlaying air. It is an interface between water and air and a boundary layer through which gases and microbes are exchanged between the two mediums (Wurl and Obbard, 2004). Even though the SML of marine environments is hostile (Agogué et al., 2005), it remains home to many microorganisms that perform various biogeochemical functions. In marine environments, the SML also accumulates anthropogenic pollutants (Wurl and Obbard, 2004). For instance, polyaromatic hydrocarbons, heavy metals, and microplastics are widespread in marine environments and can accumulate more in the SML than in the corresponding underlying water (Cincinelli et al., 2005; Cuong et al., 2008; Goßmann et al., 2023). In addition, sea foams, irregular patches formed through the accumulation of surface-active substances on sea surfaces (Rahlff et al., 2021), are a hot spot of anthropogenic pollutants (Schilling and Zessner, 2011).

This study is interested in the occurrence, spatial distribution, vertical distribution, and risk assessment of pharmaceutical compounds and other emerging contaminants (UV filters and artificial sweeteners) in the southern North Sea, strongly influenced by anthropogenic pollutants, including pharmaceutical compounds (Saux Picart et al., 2015). Anthropogenic pollutants such as heavy metals (Gillard et al., 2019), microplastics (Roscher et al., 2021), pharmaceuticals, and personal care products (Brumovský et al., 2016) are released into the North Sea. With 260,000 ship movements per annum (Unger et al., 2022), the North Sea receives further anthropogenic inputs from ships, WWTP, rivers, and other sources (Saliba et al., 2022). According to Brumovský et al. (2016), four pharmaceutical compounds (carbamazepine, atenolol, ibuprofen, and ketoprofen) and two artificial sweeteners (acesulfame and saccharin) were found to be widespread in the English Channel and Skagerrak strait which are parts of the North Sea. Pharmaceutical micropollutants from Dutch sewages and rivers usually reach the North Sea (Walraven and Laane, 2008). Since more population will reside close to the coast in the next two decades (Gaw et al., 2014), the North Sea may be prone to higher anthropogenic influx (Saux Picart et al., 2015).

The accumulation of pollutants in the SML was studied to affect the development of seafood (Price, 1998) and echinoderms (Fowler and Price, 2022) adversely. Fish eggs, larvae, foraminiferans, chaetognaths, crustaceans, snails, and jellyfish migrate to SML during their nursery and developmental periods (Gallardo et al., 2021). We previously described the possible effects of the accumulation of antibiotics in the SML of marine environments on fish development (Adenaya et al., 2023). We proposed that the consistent disposal of pharmaceutical compounds in marine environments can cause their accumulation in the SML (Adenaya et al., 2023). Consequently, we investigated the occurrence, spatial, and vertical distributions of 62 pharmaceutical compounds, one UV filter, and six artificial sweeteners in four stations along a transect from coastal areas to the open German Bight in the southern North Sea. This study is useful in assessing the vertical distribution of pharmaceutical compounds and other emerging contaminants in other aquatic environments and encourages efforts to protect the integrity of marine ecosystems and public health.

2. Materials and methods

2.1. Sampling site and sample collections

We investigated the occurrence, spatial, and vertical distributions of 62 pharmaceutical compounds, one UV filter, and six artificial sweeteners from the SML to five different depths (0 cm, 20 cm, 50 cm, 100 cm, and 150 cm) of the corresponding underlying waters in four stations in the southern North Sea (Fig. 1). One station is a coastal area (Jade Bay), one station is the estuary of the River Weser (Weser estuary) close to the city Bremerhaven, and the other two stations (NS_7 and NS_8) are in the open German Bight. Throughout this publication, we will refer to each environment with the name in parentheses. We collected seawater samples during a sampling campaign with the RV Heinke (HE 614) in March 2023.

The Jade Bay is a tidal bay influenced by the city of Wilhelmshaven, one of Germany's most prominent shipping ports. Furthermore, WWTP, hospitals, recreational centers, and Wilhelmshaven's south beach surround Jade Bay. Thus, pharmaceutical compounds originating from these sources could end up in the bay and eventually in the North Sea. Therefore, we collected seven seawater samples from Jade Bay. First, since sea foams are a hot spot for anthropogenic pollutants, we collected one foam sample (500 mL) from this coastal environment. In addition, we obtained 500 mL of SML samples and underlying water starting from 0 cm to 150 cm. The sea surface at 0 cm represents the layer of seawater immediately beneath the SML and extends a few centimeters below the surface (Chen et al., 2022).

The Weser estuary in Bremerhaven links the River Weser to the southern North Sea. Bremerhaven is also one of the prominent shipping ports in Germany with more extensive shipping activities than Wilhelmshaven harbor. Wilhelmshaven and Bremerhaven shipping ports have handled various types of cargo, including containers, bulk goods, and oil-related products (Butt et al., 2009). Aside from the prominent shipping activities, the estuary in Bremerhaven is exposed to industrial effluents and dredging (Calmano et al., 1982). River Weser discharges to the southern North Sea through the estuary at Bremerhaven. The region is characterized by agricultural production (Lange et al., 2008). It passes through industrialized and densely populated areas before linking to the North Sea (Roscher et al., 2021). For this reason, we collected six seawater samples from the estuary, starting from the SML to the underlying water from 0 cm down to 150 cm.

Unlike Jade Bay and the Weser estuary, which are semi-enclosed coastal environments, the other two stations, NS_7 and NS_8, are in the open waters (Fig. 1) with efficient water circulation due to strong currents and large surface areas that ensure adequate dilution and removal of pollutants (Laane et al., 2013). We expected that samples collected from these two stations would contain lower concentrations of pharmaceutical compounds and other micropollutants. We collected seawater samples from the SML to 150 cm at these stations.

We collected 24 seawater samples from all stations with new amber high-density polyethylene (HDPE) bottles rinsed multiple times with high-quality ethanol and MQ water to avoid sample contamination. Foam and SML samples were collected using the glass plate method described by Harvey and Burzell (1972). The glass plate was cleaned

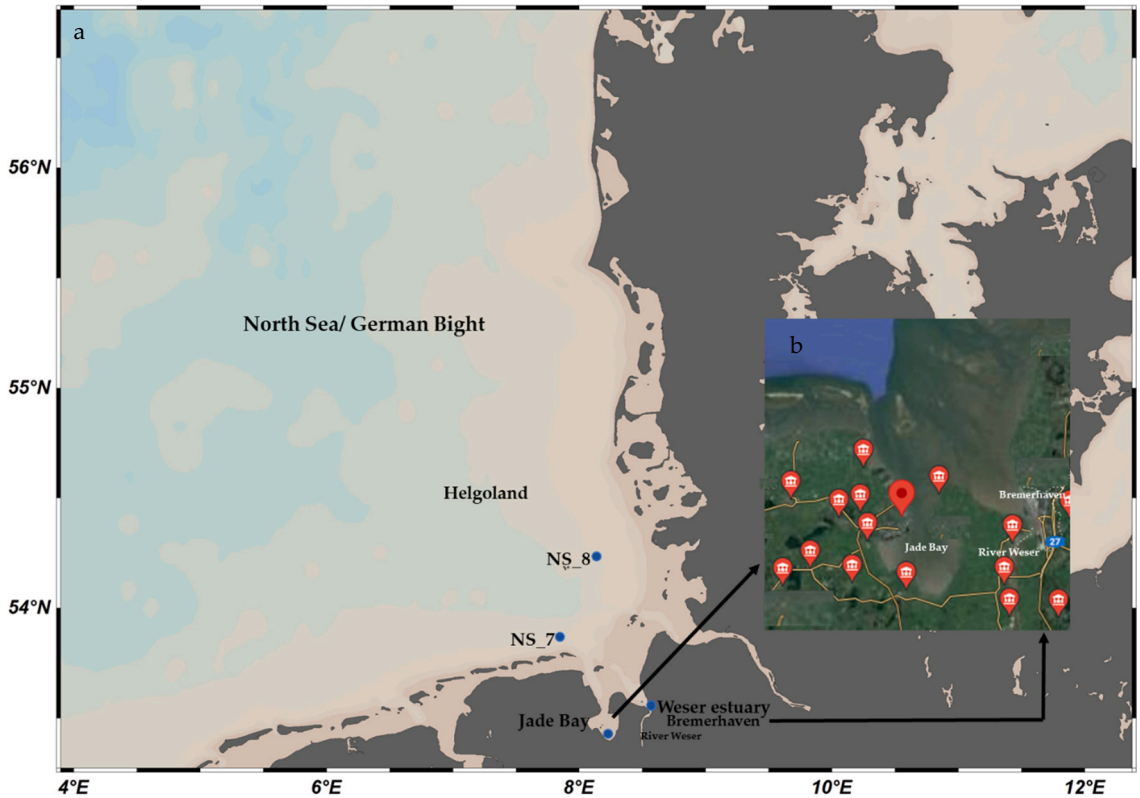

Fig. 1. (a) Map of the German Bight showing our sampling stations marked with blue dots. (b) Wastewater treatment plants (in red) near Jade Bay and the Weser estuary. Maps were generated using Schlitzer, Reiner's Ocean Data View (https://odv.awi.de, 2023) and https://www.google.de. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

multiple times with high-grade ethanol (90 %) and rinsed by dipping it into the seawater before sample collection. Seawater samples from 0 to 150 cm were collected using a van Dorn sampler. We froze each sample immediately at −20 °C after collection and at −80 °C after the cruise before sample analysis to prevent microbial degradation (Zaidi and Imam, 1999). Temperature and salinity (Table 1) were measured with the ship's Thermosalinograph. The RV HE weather station recorded wind speed.

2.2. Analysis of target compounds

All solvents used for the chemical analysis were of high chromatography quality. Methanol HPLC was purchased from Avantor (Poland), and methanol LC-MS and water HPLC were purchased from Scharlau (Barcelona, Spain). Milli-Q water and high-quality ethanol were used for all cleaning procedures in our study. Information on target compounds is presented in Table S1. In analyzing micropollutants in a complex matrix such as seawater, various compounds, and salts can interfere with the accurate quantification of target compounds. In addition, the sample preparation method, extraction method, chromatography conditions, calibration methods, and cleaning procedure can all contribute to matrix effects, affecting the reliability of the quantification analysis (Williams et al., 2023). To overcome this issue, we used isotopically labeled standards (Table S1) to account for compound loss during sample preparation and for instrument response variations due to matrix effects.

Table 1
Sampling dates and stations, temperature, salinity, wind speed, and number of samples collected at each sampling site.

Date	Sample station	Temperature (°C)	Salinity	Wind speed (m s^{-1})	Numbers of sample
12/03/2023	Jade Bay	4	30	7	7
13/03/2023	Weser estuary	6	29	10	6
17/03/2023	NS_7	6	31	8	6
19/03/2023	NS_8	6	33	8	5

First, we extracted target compounds (62 pharmaceutical compounds, one UV filter, and six artificial sweeteners) from our samples using the solid phase extraction (SPE) method. We used 200 mg Oasis HLB cartridges supplied by Waters (Milford, USA) for SPE. Filtration of 500 mL of each sample was done through a 1 μm glass microfiber filter (47 mm; Whatman GFB Cytiva, China) with an adjusted pH of 2 (Baena-Nogueras et al., 2016a, 2016b; Baena-Nogueras et al., 2018). To validate the extraction method, we spiked each sample with 50 μL of isotopically labeled surrogate compounds, i.e., ibuprofen-D_3 and carbamazepine-D_{10}. Isotopically labeled surrogates enhance the accuracy and reliability of quantitative measurements of the target compounds in seawater samples (Williams et al., 2023).

We then loaded six samples at a time onto the SPE instrument (Dionex™ AutoTrace™ 280 ThermoFisher, USA). The instrument was conditioned to start the extraction procedure by activating the HLB cartridges with 10 mL of methanol HPLC and 10 mL of water HPLC. Each sample was passed through the SPE columns at 6 mL min^{-1}. The extracted sorbent was washed twice with 10 mL of water HPLC and air-dried for 10 min. Extracts were eluted into clean tubes with 10 mL of methanol (LC-MS), evaporating under nitrogen flow within 1 h at 42 °C. Extracts were resuspended in a mixture of methanol and water mixture (50:50 v.v), sonicated for 5 min at room temperature, and finally filtered through 0.22 μm polytetrafluoroethylene (PTFE) filters (Teknochroma, Spain) before analysis. Sixteen isotopically labeled compounds were

then added at 50 μg L^{-1} as internal standards to account for matrix effects during mass spectrometry detection.

We used the protocols developed by Baena-Nogueras et al. (2016a, 2016b) and Baena-Nogueras et al. (2018) for the same pharmaceutical compounds and other micropollutants i.e. UV filter and artificial sweeteners for the analysis of target compounds in our samples. Chemical analysis of target compounds was carried out using ultra-performance liquid chromatography - triple quadrupole mass spectrometry in tandem with ionization spray (UPLC-QqQMS/MS) (Bruker EVOQ Elite system Bruker, USA). Before sample analysis, the column was sequentially washed with 100 % methanol and water to eliminate the possible matrix effect. The chromatographic separations of pharmaceutical compounds, UV filters, and artificial sweeteners were performed on a C_{18} Intensity Solo UPLC Column (2.1 mm × 100 mm, 1.8 μm pore size; Acquity, Waters, Ireland). Chromatographic separation was obtained under positive and negative electrospray modes (ESI+/−). For the chromatographic separation of target compounds in positive electrospray mode, the sequential aqueous phase was LC water with 10 mM of formic acid (95 %) and methanol with 10 mM of formic acid (5 %) (pH = 3) and the organic phase was 100 % methanol. For the chromatographic separation of target compounds in negative electrospray mode, we used LC water with 5 mM ammonia (95 %), methanol, and 5 mM ammonia (5 %) (pH = 8) as the aqueous phase. 100 % methanol was used as the organic phase.

Chromatographic separation under positive and negative electrospray modes is a crucial step in mass spectrometry to create ions and expand the range of detectable micropollutants (Medhe, 2018). Positive electrospray mode is commonly used for compounds that readily lose electrons, while negative electrospray mode is often used for compounds with a high electron affinity (Medhe, 2018). For micropollutants in positive ionization electrospray mode, the percentage of methanol linearly increased from 0 % to 100 % for 5 min and was maintained at 100 % for 3 min. The injection volume of samples and standards was 10 μL and system run time per sample was 10 min, with a 2-min re-equilibration period. For micropollutants in negative electrospray mode, the percentage of methanol linearly increased from 0 % to 100 % for 2 min and maintained for 4 min. The injection volume of samples and standards was also 10 μL, and the system run time per sample was 8 min, including a 2-min re-equilibration period. In both cases, the flow rate was 0.4 mL min^{-1}, the pressure was 0.4 bar, and the temperature was 45 °C.

Information regarding the mass spectrometry (MS) parameter setting is presented in Table S2. The concentration of internal standards used for calibrations ranged from 1 to 100 μg L^{-1}. We identified target compounds in our samples by matching their retention times and signal intensities to those of the standards and by the ratio of two multiple reaction monitoring (MRM) transitions. We calculated the limits of detection (LOD) and limit of quantification (LOQ) using the signal-to-noise ratio of 3 and 10, respectively (Baena-Nogueras et al., 2016a, 2016b; Baena-Nogueras et al., 2018). Recoveries of target compounds were previously assessed in other manuscripts (Baena-Nogueras et al., 2016a, 2016b; Baena-Nogueras et al., 2018). Identification and quantification of target compounds were obtained with the Bruker MS Workstation (V8.2.RC8). We performed graphical representations of our results with the SPSS Statistics V28.0.1.0 (142) software package.

2.3. Environmental risk assessment

Based on available ecotoxicological data for marine species, we performed a risk assessment of 19 pharmaceutical compounds detected in our study for marine organisms belonging to two trophic levels – algae (*Skeletonema marinoi*, *Anabaena* sp., *Synechococcus leopoliensis*, and *Cyclotella meneghiniana*) and invertebrates (*Artemia salina*, *Mytilus* sp. and *Navicula ramosissima*). Due to abundant data for freshwater organisms, we also conducted an environmental risk assessment of detected pharmaceutical compounds, UV filter, and artificial sweeteners utilizing ecotoxicological data for freshwater organisms from three different trophic levels – algae (*Anabaena cylindrica*, *Chlorella* sp., *Desmodesmus subspicatus*, *Scenedesmus subspicatus*, *Raphidocelis subcapitata*, and *Cyanobacteria* spp.), invertebrates (*Ceriodaphnia dubia*, *Cladocerans* sp., *Daphnia magna*, *Hydra attenuata*, *Isochrysis galbana*, and *Thamnocephalus platyurus*), and fish (*Brachydanio rerio*, *Danio rerio*, *Oryzias latipes*, *Pimephales promelas*, *Siriella armata*, and *Xenopus* sp.). The risk assessment followed the worst-case scenario described by Biel-Maeso et al. (2018).

To assess the environmental risk of target compounds in each sampling station, we calculated risk quotients (RQ) by comparing the measured environmental concentrations (MEC) of the detected compounds to predicted no-effect concentrations (PNEC) as described by Okonski et al. (2021). We used the highest concentration of each detected compound in our samples for MEC (Biel-Maeso et al., 2018). PNEC was estimated by dividing the lowest EC_{50} (half maximal effective concentration) values, which represent the concentration at which the detected compound results in 50 % mortality of marine and freshwater organisms (ECB, 2003), by an assessment factor (AF) of 1000 (Biel-Maeso et al., 2018). We obtained EC_{50} values for the detected compounds from the literature (see Tables S3 and S4). The overall RQ was then calculated using equations (a) and (b).

$$PNEC = \frac{E_{50}}{AF} \quad (a)$$

$$RQ = \frac{MEC}{PNEC} \quad (b)$$

3. Results and discussion

3.1. Occurrence and spatial distributions of micropollutants

To our knowledge, this is the first study to examine the occurrence and vertical distribution of micropollutants of emerging concern (pharmaceutical compounds, UV filters, and artificial sweeteners) in the SML and near-surface layer in the southern North Sea. Although numerous surface waters in Germany are safeguarded and considered free from chemical and biological pollutants (Wolfram et al., 2021; Cara et al., 2022), our study found that among the 62 pharmaceutical compounds examined, 33, 23, 26, and 25 were detected in all seawater samples collected from the Jade Bay, the Weser estuary, NS_7, and NS_8, respectively. We also detected one UV filter in each station, four artificial sweeteners in Jade Bay, five in Weser estuary, and three in other sampling stations (Table 2).

The average concentrations of pharmaceutical compounds, UV filter, and artificial sweeteners measured differed considerably by the station (Fig. 2). The average concentrations of pharmaceutical compounds in Jade Bay, the Weser estuary, NS_7, and NS_8 are 33.32 ng L^{-1}, 31.94 ng L^{-1}, 17.10 ng L^{-1}, and 19.09 ng L^{-1}, respectively. The concentration of pharmaceutical compounds found in the Jade Bay and the Weser estuary indicates a likely source of direct wastewater input from nearby WWTP (Fig. 1). Since the Weser estuary also receives freshwater inputs directly from the River Weser, which flows through several cities including Bremen, introducing pharmaceutical compounds into the environment is also expected. The measured pharmaceutical concentration in both Jade Bay and the Weser estuary could also result from port activities and dredging (Lompe and Nurlu-Bruns, 2012). The Weser estuary also has the highest concentration of UV filter (162.10 ng L^{-1}), followed by the Jade Bay (21.65 ng L^{-1}).

In NS_7 and NS_8, the highest concentrations of pharmaceutical compounds were 143.12 ng L^{-1} and 105.32 ng L^{-1}, respectively. The concentrations of UV filter in both stations were 0.43 ng L^{-1} and 6.23 ng L^{-1}. Aside from concentration differences, most of the pharmaceutical compounds detected in Jade Bay were also present in the two stations in the open German Bight (Table 2). The strong tidal current in the North Sea may further distribute coastal pollutants to the open North Sea.

Table 2
Concentration ranges of pharmaceutical compounds, UV filter, and artificial sweeteners detected in the southern North Sea. DF detection frequency of each pharmaceutical compound in each environment, n.d = undetected.

Micropollutant groups	Compound	Jade Bay		Weser estuary		NS_7		NS_8	
		Range (ng L^{-1})	DF (%)	Range (ng L^{-1})	DF (%)	Range (ng L^{-1})	DF (%)	Range (ng L^{-1})	DF (%)
Anti-inflammatories	Acetaminophen	n.d	0	n.d	0	n.d	0	n.d	0
	Diclofenac	n.d – 3.51	57	2.95–4.92	100	n.d – 1.39	17	n.d – 1.29	20
	Fenoprofen	n.d – 18.77	29	n.d	0	n.d	0	n.d –	0
	Ibuprofen	9.2–52.07	100	36.57–51.94	100	10.80–15.82	100	1.23–15.48	100
	Indomethacin	n.d – 4.30	57	1.99–3.61	100	n.d – 0.81	33	n.d – 0.50	20
	Ketoprofen	n.d – 21.90	100	n.d	0	n.d	0	n.d	0
	Mefenamic acid	n.d – 7.29	71	0.05–1.11	100	n.d – 1.72	33	n.d – 0.20	20
	Naproxen	n.d – 169.75	43	n.d – 1.87	**83**	n.d	0	n.d	0
	Phenylbutazone	Nd	0	n.d	0	n.d	0	n.d	0
	Salicylic acid	18.08–201.60	100	7.43–45.39	100	10.10–127.30	100	26.07–87.86	100
Beta-blockers	Atenolol	n.d	0	n.d	0	n.d – 1.93	17	n.d – 1.36	40
	Pindolol	n.d	0	n.d	0	n.d	0	n.d	0
	Nadolol	n.d – 1.31	43	n.d	0	n.d	0	n.d	0
	Metoprolol	1.56–2.72	100	10.89–21.23	100	1.61–2.26	100	0.70–0.89	100
	Timolol	n.d	0	n.d	0	n.d	0	n.d	0
	Propranolol	n.d	0	2.22–2.23	100	n.d – 2.90	17	n.d	0
Lipid-regulators	Bezafibrate	0.26–0.44	100	0.80–1.48	100	n.d – 0.16	17	0.13–0.21	100
	Clofibric acid	n.d	0	n.d	0	n.d	0	n.d	0
	Fenofibrate	0.71–3.90	100	n.d – 2.09	83	n.d – 0.18	83	0.43–1.48	100
	Gemfibrozil	0.10–0.24	100	0.11–0.15	100	0.09–0.18	100	n.d –	100
	Pravastin	n.d	0	n.d	0	n.d	0	n.d	0
Diuretics	Furosemide	n.d	0	n.d – 2.65	17	n.d	0	n.d	0
	Hydrochlorothiazide	1.77–3.41	100	6.30–9.53	100	n.d – 0.79	66	0.86–1.06	100
Stimulants	Caffeine	4.17–97.87	100	52.41–102.31	100	5.13–52.54	100	4.03–100.09	100
	Carbamazepine	2.94–5.11	100	9.91–18.67	100	3.21–3.70	100	1.62–1.86	100
	Fluoxetine	n.d – 0.14	43	0.07–0.11	100	n.d – 1.59	50	1.39–1.49	100
Anti-depressants	Amitriptyline	n.d	0	n.d	0	n.d	0	n.d	0
Histamine-receptor antagonist	Ranitidine	n.d	0	n.d	0	n.d	0	n.d	0
	Famotidine	n.d	0	n.d	0	n.d – 0.46	33	n.d – 0.38	40
Lincosamides	Lincomycin	n.d – 0.17	14	n.d	0	n.d	0	n.d	0
	Clindamycin	0.70–1.57	100	2.14–5.09	100	0.33–0.71	100	n.d – 1.37	60
Macrolides	Spiramycin	n.d	0	n.d	0	n.d	0	n.d	0
	Azithromycin	n.d	0	n.d	0	n.d	0	n.d	0
	Clarithromycin	1.72–4.32	100	1.55–1.75	100	0.70–1.79	100	1.22–1.30	100
	Tylosin	0.75–2.89	100	n.d – 1.28	50	0.60–3.33	100	0.84–1.33	100
	Erythromycin	n.d – 11.38	57	n.d	0	3.39–5.26	100	n.d – 9.34	80
	Roxithromycin	n.d	0	n.d	0	n.d	0	n.d	0
	Azithromycin	n.d	0	n.d	0	n.d	0	n.d	0
Quinolones	Ofloxacin	n.d – 7.17	58	n.d	0	n.d – 5.43	66	n.d	0
	Enrofloxacin	n.d	0	n.d	0	n.d	0	n.d	0
	Danofloxacin	n.d	0	n.d	0	n.d – 11.14	33	n.d – 10.31	60
	Sparfloxacin	n.d	0	n.d	0	n.d	0	n.d –	0
	Flumequine	n.d	0	n.d	0	n.d	0	n.d –	0
Nitroimidazols	Metronidazole	n.d – 0.39	29	n.d	0	n.d	0	n.d –	0
	Ornidazole	n.d – 45.91	14	n.d	0	n.d	0	n.d – 2.04	20
Phenicol	Chloramphenicol	n.d – 0.79	29	n.d	0	n.d – 2.98	17	n.d	0
Sulfonamides	Sulfaguanidine	n.d	0	n.d	0	n.d	0	n.d	0
	Sulfadiazine	n.d – 22.22	14	n.d	0	n.d	0	n.d	0
	Sulfamethizole	n.d	0	n.d	0	n.d	0	n.d	0
	Sulfamethazine	n.d	0	n.d	0	n.d	0	n.d	0
	Sulfamethoxazole	n.d – 0.88	43	0.40–1.02	100	n.d	0	n.d – 0.48	80
	Sulfathiazole	n.d	0	n.d	0	n.d	0	n.d	0
	Sulfisoxazole	n.d	0	n.d	0	n.d	0	n.d	0
	Sulfadimethoxine	n.d	0	n.d	0	n.d	0	n.d	0
Aminocoumarin	Novobiocin	6.42–50.77	100	6.37–15.18	100	4.04–9.37	100	6.95–25.96	100
Tetracyclines	Doxycycline	n.d – 20.18	57	n.d	0	n.d	0	n.d – 9.25	20
	Chlortetracycline	n.d	0	n.d	0	n.d	0	n.d	0
	Oxytetracycline	n.d – 4.50	14	n.d	0	n.d	0	n.d	0
Diaminopyrimidines	Trimethoprim	n.d – 2.39	86	0.82–1.13	100	0.50–1.29	100	0.72–1.05	100
Rifampicin	Rifampicin	n.d	0	n.d	0	n.d	0	n.d	0
Other pharmaceuticals	Methotrexate	n.d	0	n.d	0	n.d	0	n.d	0
	Glibenclamide	n.d – 7.16	71	n.d – 32.88	**83**	n.d – 13.16	83	n.d – 6.35	40
UV-filter	Benzophenone-4	11.07–56.57	100	89.16–189.82	100	n.d – 1.93	33	1.59–10.48	100
Artificial sweeteners	Acesulfame	9.46–19.15	100	23.17–34.41	100	3.16–7.54	100	6.93–10.64	100
	Saccharin	13.88–50.32	100	60.00–80.50	100	4.21–12.16	100	11.64–25.10	100
	Cyclamate	n.d – 212.00	57	4.32–65.50	100	n.d	0	n.d	0
	Aspartame	n.d	0	n.d – 1.38	50	n.d	0	n.d	0
	Sucralose	705.03–3218.35	100	700.73–1014.00	100	1255.75–2001.31	100	649.61–1866.34	100
	Neohesperidin dihydrochalcone	n.d	0	n.d	0	n.d	0	n.d	0

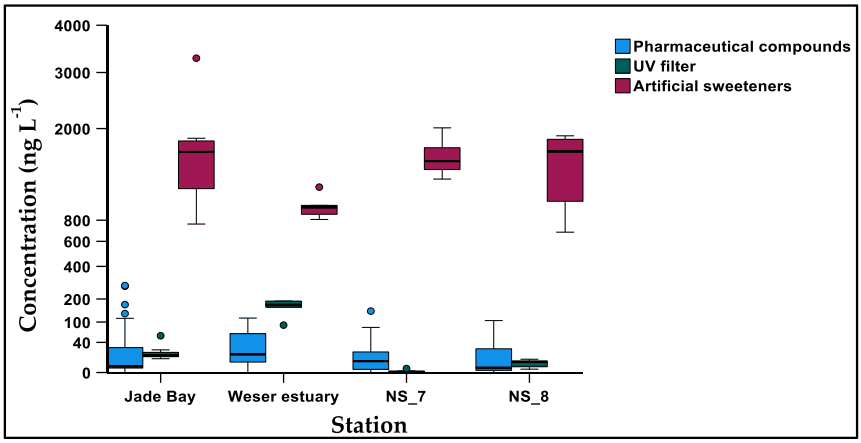

Fig. 2. Concentrations of pharmaceutical compounds, UV filter, and artificial sweeteners in the Jade Bay, the Weser estuary, NS_7 and NS_8. The y-axis is presented on a logarithmic scale (base 10) to facilitate the visibility of lower and higher concentrations.

However, factors such as dilution, turbulence, and the low retention time of pollutants in the open sea contribute to their lower concentrations in open coastal waters (Stoderegger and Herndl, 1999; Lara-Martín et al., 2014). Furthermore, the average concentrations of artificial sweeteners detected in Jade Bay, NS_7, and NS_8 were higher than those measured in the Weser estuary (Fig. 2). This could be attributed to the proximity of Jade Bay to the open southern North Sea. Artificial sweetener concentrations exhibit a sequence of proximity from Jade Bay (average = 1666.60 ng L^{-1}) to NS_7 (average = 1580.00 ng L^{-1}) and finally to NS_8 (average = 1410.80 ng L^{-1}), probably due to tidal transportation influencing their distributions from the coastal area to the open waters (Tong et al., 2022).

3.2. Spatial variability and detection frequency of micropollutants

The distribution of pharmaceutical compounds in marine environments depends on the locations of sources and transport processes because different activities are performed around different coastal areas (Alygizakis et al., 2016). This is also the case in our study because a few compounds (with high detection frequency) occurred in certain stations and not others. An example of such compounds is ofloxacin, which is detected in Jade Bay and NS_7 but not in the Weser estuary and NS_8. In addition, ketoprofen and propanol were detected only in the Jade Bay and the Weser estuary, respectively, but not in NS_7 and NS_8. Furthermore, the artificial sweetener cyclamate was detected in the Jade Bay and the Weser estuary, while aspartame was detected only in the Weser estuary (see Table 2).

All pharmaceutical compounds detected in all stations belong to seven different pharmaceutical groups, but the concentration of each pharmaceutical group varied with the sampling stations (Fig. S1). In Jade Bay, NS_7, and NS_8, the total concentrations of anti-inflammatory compounds (ibuprofen and salicylic acid, metoprolol, and gemfibrozil) were higher than the total concentrations of other pharmaceutical groups. The concentration of salicylic acid was the highest, ranging from 18.08 to 201.60 ng L^{-1} in Jade Bay, 7.43–45.39 ng L^{-1} in Weser estuary, 10.10–127.30 ng L^{-1} in NS_7 and 26.07–87.86 ng L^{-1} in NS_8. At the same time, stimulants (caffeine, carbamazepine, and fluoxetine) have the highest concentrations in the Weser estuary (average = 92.20 ng L^{-1}) compared to Jade Bay. The concentration of caffeine was also the highest among this pharmaceutical group. Caffeine concentrations ranged from 4.17 to 97.87 ng L^{-1} in Jade Bay, 52.41–102.31 ng L^{-1} in Weser estuary, 5.13–52.54 ng L^{-1} in NS_7, and 4.3–100.09 ng L^{-1} in NS_8. Lastly, the average concentration of antibiotics (clarithromycin, clindamycin, erythromycin, novobiocin, ofloxacin, trimethoprim, and tylosin) was higher in Jade Bay (43.98 ng L^{-1}). Indeed, Jade Bay has the highest antibiotic concentrations than Weser estuary (18.40 ng L^{-1}), NS_7 (21.30 ng L^{-1}), and NS_8 (30.30 ng L^{-1}), which indicates the direct discharge of pharmaceutical compounds into the coastal region via municipal WWTP.

Of all the artificial sweeteners detected in our study, sucralose had the highest concentration, followed by saccharin. The concentration of sucralose ranged from 705.03 to 3218.35 ng L^{-1} in Jade Bay and 700.73–1014.00 ng L^{-1} in Weser estuary. Some artificial sweeteners, such as sucralose and saccharine, remain unaffected by WWTP processes before being introduced into coastal ecosystems (Brorström-Lundén et al., 2008). This may, therefore, account for the high concentrations of artificial sweeteners observed in our study. Artificial sweeteners are stable and non-evaporating. They are mainly used as food additives and agents for pharmaceutical production and are good tracers of urban wastewater discharge into marine environments (Buerge et al., 2009).

Among the 62 pharmaceutical compounds, we detected caffeine, carbamazepine, gemfibrozil, ibuprofen, metoprolol, and salicylic acid in all our samples, i.e., 100 % detection frequency (Table 2). In contrast, bezafibrate, cyclamate, diclofenac, fenofibrate, fluoxetine, glibenclamide, hydrochlorothiazide, indomethacin, mefenamic acid, have a detection frequency $\geq$ 50 %. Antibiotics such as clarithromycin and novobiocin have 100 % detection frequency, while clindamycin, trimethoprim, and tylosin have >90 % detection frequency. Erythromycin and sulfamethoxazole were detected in 58 % and 54 % of all analyzed samples in our study. We also detected ofloxacin, doxycycline, and lincomycin with detection frequencies >33 %. One UV filter (Benzophenone-4) was detected in 83 % of our samples, while three artificial sweeteners (acesulfame, saccharin, and sucralose) had 100 % detection frequency.

The detection of caffeine, carbamazepine, gemfibrozil, ibuprofen, and salicylic acid in all seawater samples is consistent with other studies (Benotti and Brownawell, 2007; Lara-Martín et al., 2014; Biel-Maeso et al., 2018; Afsa et al., 2020). According to Weigel et al. (2002), caffeine was present throughout the North Sea. Acesulfame, carbamazepine, ibuprofen, and sucralose were also detected in over 90 % of seawater samples obtained from the English Channel and Skagerrak Strait of the North Sea (Brumovský et al., 2016). The widespread consumption of acesulfame, caffeine, carbamazepine, ibuprofen, and sucralose, and subsequent excretion in wastewater, led to their use as markers for micropollutants in marine environments directly affected by WWTP. For instance, caffeine is a widely consumed compound found primarily in coffee, soft drinks, and tea. As a result, it is often detected in wastewater and seawater samples (Tritsch et al., 2022). Similarly, ibuprofen, a common over-the-counter medication, is widely used by the general population to relieve pain and inflammation. This also results in its presence in wastewater and, finally, in receiving marine environments (Jan-Roblero and Cruz-Maya, 2023).

Clarithromycin, clindamycin, ofloxacin, trimethoprim, and sulfamethoxazole consumed by a local population in Kenzening (Germany) were found to be widespread in municipal wastewater influents and effluents (Alexy et al., 2006). Clarithromycin and erythromycin were detected in different parts of the German Bight (Ondruch et al., 2023). Indeed, sulfamethoxazole was detected in 50 % of seawater samples obtained in the English Channel and Skagerrak Strait of the North Sea (Brumovský et al., 2016). Inputs from hospitals, agricultural, and WWTP effluents are the major driving forces of these antibiotics in coastal waters (Zhang et al., 2013; Lin et al., 2023). The dilution effect has been effective in reducing the concentrations of antibiotics that have been detected in coastal waters.

The removal rate of these pharmaceutical compounds in municipal WWTP varied considerably due to their different physical and chemical properties (Verlicchi et al., 2012). As such, some of them are recalcitrant and persistent in marine environments (Alexy et al., 2006), contributing to their consistent detection in coastal waters globally (Yamamoto et al., 2009). Caffeine, carbamazepine, and ibuprofen are used as a proxy for pharmaceutical contamination of marine environments and waters surrounded by WWTP (Lara-Martín et al., 2014). This is because their removal rates by WWTP are usually <70 % (Lara-Martín et al., 2014; Tiwari et al., 2017). Erythromycin, ofloxacin, trimethoprim, and sulfamethoxazole are widely distributed because WWTP does not completely remove them (Alexy et al., 2006). Additionally, the antibiotics are not easily degraded and can remain active for >100 days in coastal waters (Benotti and Brownawell, 2009).

3.3. Vertical distribution of micropollutants

We investigated the vertical distribution of micropollutants from the SML down to 150 cm depth in the southern North Sea. In Jade Bay, we included one foam sample that could potentially accumulate anthropogenic pollutants (Pierce et al., 2003). Indeed, there were no differences in the total numbers of pharmaceutical compounds detected in our foam sample, the SML sample, and seawater samples collected at 0 cm and 150 cm depth (Fig. S2). The total number of compounds detected at these depths was 30, 29, 30, and 28, respectively. However, the number of detected compounds in seawater samples collected at 20 cm, 50 cm, and 100 cm slightly differed and were 25, 22, and 23, respectively. Due to a lower salinity, water from WWTP is less dense than seawater and tends to float and accumulate in surface seawater (Lara-Martín et al., 2014). Thus, high numbers of pharmaceutical compounds detected in the SML and near the surface seawater were expected.

The total concentrations of pharmaceutical compounds measured in Jade Bay varied considerably with depths (Fig. 3a). The SML had the highest concentration of pharmaceutical compounds (422.13 ng L^{-1}) compared to the foam sample (389.90 ng L^{-1}) and samples from other depths. The SML also had the highest concentration of UV filters (56.56 ng L^{-1}), while the foam had the highest concentration of artificial sweeteners (3287.81 ng L^{-1}) (Fig. S3). Although more compounds were detected at a depth of 0 cm compared to 20 cm, the total concentration of pharmaceutical compounds measured was lower at 0 cm (197.75 ng L^{-1}) than at 20 cm (336.58 ng L^{-1}) in Jade Bay. The lowest concentration of pharmaceutical compounds was measured at a depth of 50 cm (96.68 ng L^{-1}). Aside from the direct discharge of pharmaceutical

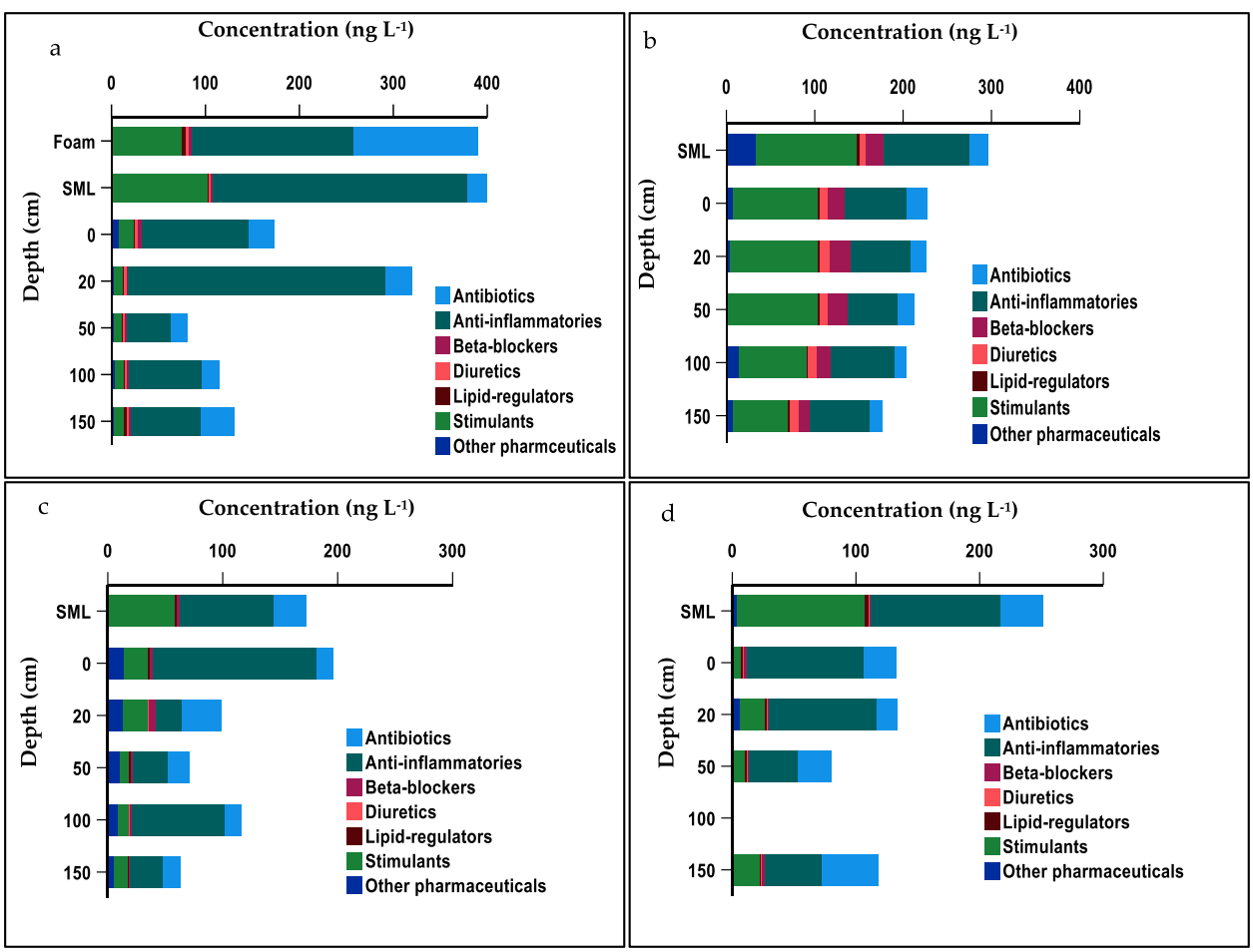

Fig. 3. Vertical distribution of pharmaceutical compounds measured in samples taken from (a) the Jade Bay, (b) Weser estuary, (c) station NS_7, and (d) station NS_8.

compounds via WWTP into Jade Bay, the coastal region is a tidal flat with high concentrations of suspended particles and high surface-active substances (surfactants) at the surface (Götschenberg and Kahlfeld, 2008). This could account for the high concentrations of pharmaceutical compounds observed. Some artificial surfactants are also discharged into coastal domains through WWTP (Lara-Martín et al., 2014). This could influence the accumulation of pollutants in the SML of this coastal area.

The total number of pharmaceutical compounds was the same in the Weser estuary, irrespective of the depth (Fig. S2). The consistent trends in the Weser estuary may be due to continuous water mixing (homogeneity) resulting from the continuous inflow and outflow of freshwater and seawater in the environment (Alygizakis et al., 2016). Nonetheless, the total concentration of pharmaceutical compounds detected in the Weser estuary was higher in the SML than in other depths (Fig. 3b), further highlighting the ability of the SML to accumulate more pollutants than the water column (Wurl and Obbard, 2004). The estuary receives pollutant inputs from WWTP in Bremerhaven, as well as agricultural and other industrialized areas surrounding the station (Lange et al., 2008). These inputs could contribute to the total concentrations of pharmaceutical compounds observed not only in the SML but also at all depths of the Weser estuary.

At the two stations in the open German Bight (NS_7 and NS_8), more pharmaceutical compounds were detected from the SML to 20 cm, and like in Jade Bay, the total concentration of pharmaceutical compounds varied with depths. At station NS_7, the total concentration of pharmaceutical compounds was higher in the sample collected at 0 cm (196.35 ng L^{-1}) and in the SML (172.71 ng L^{-1}) (Fig. 3c). In NS_8, the total concentration of pharmaceutical compounds in the SML was the highest (Fig. 3d). The total concentrations of pharmaceutical compounds measured in seawater samples collected at 50 cm depth in NS_7 (70.97 ng L^{-1}) and NS_8 (81.98 ng L^{-1}) was the lowest, like the trend observed in Jade Bay. Coastal regions are hydrodynamically complex (Laignel et al., 2023). Further research is therefore needed to investigate and understand the hydrodynamic factors that could contribute to the low concentrations of pharmaceutical compounds at 50 cm depth in these coastal areas.

The effect of wind on the vertical distributions of the total concentrations of pharmaceutical compounds at each sampling depth must be considered because wind creates turbulence that transports pollutants from the water column to the surface (Gioia et al., 2011). Wind-driven surface currents transport anthropogenic pollutants present in the seawater surface (Ahrens, 2011). High wind speed was recorded when each sample was collected from each sampling station (see Table 1). High winds drive surface water movement, potentially redistributing pharmaceutical compounds to different areas within marine environments (Tong et al., 2022). The depths 0 to 150 cm represent the shallowest part of water columns where wind, in most cases, induces the movement and distribution of nutrients in the North Sea (Wiltshire et al., 2010).

As a natural surfactant film, the SML is formed through biological activities at the water-air interface and is typically composed of hydrophobic moieties. The hydrophobic nature of the film, therefore, favors the accumulation of pollutants on sea surfaces (Wurl and Holmes, 2008). Moreover, about 60 % of pharmaceutical compounds are hydrophobic (Hogarth et al., 2021) and this was studied to hinder their adsorption behaviors in freshwater environments (Wu et al., 2016). Indeed, many pharmaceutical drugs contain high amounts of surface-active excipients (van Hoogevest et al., 2011) to enhance their solubility in water. These excipients possess surfactant properties, which also accumulate on water surfaces. The accumulation of pharmaceutical compounds in the SML is therefore expected.

Adenaya et al. (2023) discussed the ecotoxicological effects of pharmaceutical compounds such as antibiotics in the SML. One of the most adverse effects is the inhibition of fish development since fish larvae and eggs reside in the SML during their developmental phases. This is because the SML significantly impacts the development and distribution of several aquatic organisms (Gallardo et al., 2021; Fowler and Price, 2022). Combining SML samples with samples from various other depths may be necessary to draw a comprehensive conclusion about the distribution and behavior of pharmaceutical compounds in coastal waters, particularly in response to anthropogenic activities. This approach can provide a more comprehensive and accurate overview of environmental contaminant concentrations. Further research is also needed to understand pollutant behavior and exchange in the water column, especially at the SML and near-surface layer of marine environments.

3.4. Vertical distribution of selected micropollutants

As mentioned in Section 3.2, caffeine, carbamazepine, gemfibrozil, ibuprofen, metoprolol, and salicylic acid, were detected in all our samples with a detection frequency of 100 %. We, therefore, selected these pharmaceutical compounds to understand their dynamics at each sampling depth in each sampling station. The antibiotics clarithromycin, clindamycin, novobiocin, trimethoprim, and tylosin with a detection frequency > 90 % were also selected for vertical distribution analysis. In addition, we selected acesulfame, sucralose, and saccharin because these artificial sweeteners were also detected in all our samples. Sucralose and saccharin also had relatively high concentrations in all sampling stations but mostly in the foam sample collected in Jade Bay. The concentrations of all pharmaceutical compounds detected at each depth are presented in Tables S5, S6, S7, and S8.

At varying depths in the Jade Bay, there were differences in the concentrations of caffeine, carbamazepine, gemfibrozil, ibuprofen, and salicylic acid (Fig. 4). In Jade Bay, caffeine was found to accumulate more in the SML (97.87 ng L^{-1}) than in the foam (68.94 ng L^{-1}). In contrast, its concentration was lower at a depth of 150 cm (5.69 ng L^{-1}). Additionally, other compounds accumulated more in the foam sample and showed no significant difference in concentration across different depths. For example, ibuprofen concentration in the foam was 52.07 ng L^{-1}. The concentration of ibuprofen measured at 0 and 150 cm depth showed almost no difference (23.71 ng L^{-1} and 31.60 ng L^{-1}).

In the Weser estuary, caffeine, and salicylic acid concentrations were higher in the SML and lower at 150 cm depths. However, the concentrations of other compounds varied with no distinct vertical distribution pattern. The compounds were characterized by a more homogenous vertical distribution in NS_7 and NS_8. Caffeine remained higher in the SML of Weser estuary, NS_7, and NS_8 compared to other depths, similar to what we observed in Jade Bay. The concentration of caffeine in the SML in the Weser estuary, NS_7, and NS_8 was 102.31 ng L^{-1}, 52.54 ng L^{-1}, and 52.53 ng L^{-1}, respectively. The vertical distribution of salicylic acid, an anti-inflammatory compound, varied greatly with sampling depth and sampling stations. Still, higher concentrations were measured from the SML down to 20 cm in all sampling stations.

Fig. S4 shows the vertical distribution of clarithromycin, clindamycin, novobiocin, trimethoprim, and tylosin. In Jade Bay, the foam sample had higher concentrations of clarithromycin, novobiocin, trimethoprim, and tylosin (4.32 ng L^{-1}, 50.77 ng L^{-1}, 2.39 ng L^{-1}, and 2.89 ng L^{-1}, respectively). The concentrations of these compounds remained relatively consistent from the SML to a depth of 150 cm, except for a slight increase in the concentration of trimethoprim (1.58 ng L^{-1}) in the SML. The Weser estuary, NS_7, and NS_8 were characterized with homogenous concentrations of clarithromycin and trimethoprim. However, clindamycin, novobiocin, and tylosin concentrations differed at each depth in these stations with no distinct vertical distribution patterns. Novobiocin and tylosin were, however, higher in the upper water surface (from 0 to 150 cm). Interestingly, a similar concentration of novobiocin was obtained in samples collected at 100 cm in the Jade Bay, NS_7, and NS_8. All sampling stations present a slight increase in novobiocin concentrations at 150 cm.

Artificial sweeteners, especially acesulfame, saccharin, and

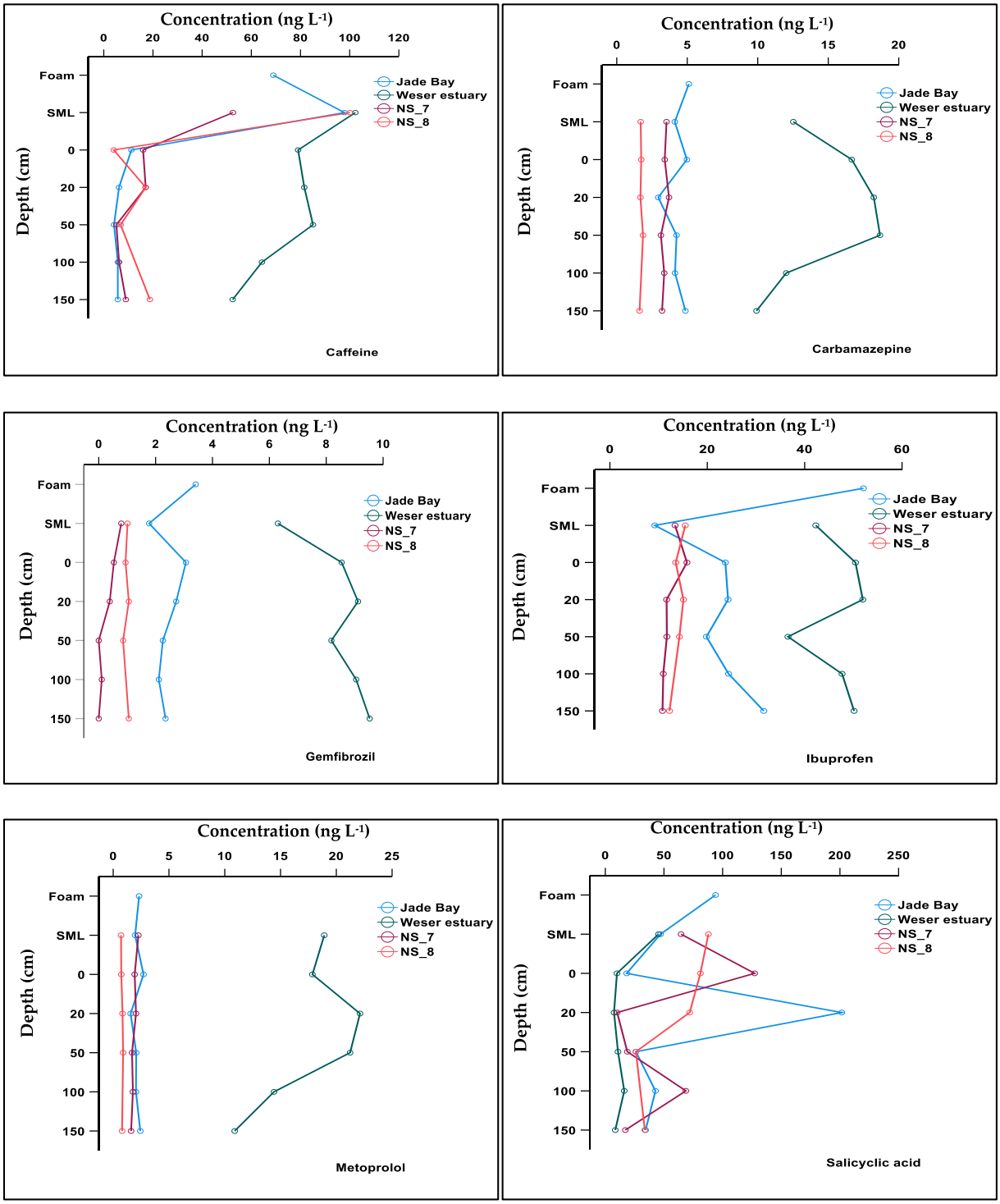

Fig. 4. Vertical distribution of caffeine, carbamazepine, gemfibrozil, ibuprofen, metoprolol, and salicylic acid in seawater samples collected in this study.

sucralose, were detected at all sampling depths, and we present their vertical distribution in Fig. S5. In Jade Bay, the concentration of acesulfame was higher in the SML (19.15 ng L^{-1}), but saccharin and sucralose had higher concentrations in the foam sample (50.32 ng L^{-1} and 3218.35 ng L^{-1}). The SML had the lowest concentration of sucralose (705.02 ng L^{-1}), while the sample collected at a depth of 150 cm had the lowest concentration of saccharin (13.87 ng L^{-1}). In the Weser estuary, the highest concentration of acesulfame was measured at a depth of 150 cm (34.40 ng L^{-1}). Saccharin and sucralose concentrations showed no difference from the SML down to 150 cm. In NS_7 and NS_8, the concentrations of acesulfame and saccharin showed no difference from the SML down to 150 cm. Sucralose concentrations, however, differed with depth. At each station, the concentration of sucralose was higher at 150 cm depth (2001.33 ng L^{-1} and 1866.33 ng L^{-1}, respectively). At station NS_8, the concentration of sucralose in the SML was the lowest, just like the trend observed in Jade Bay.

3.5. Environmental risk assessment

Utilizing the ecotoxicological information available for marine

organisms, we conducted a risk evaluation of 19 pharmaceutical compounds identified in our study for marine organisms from two trophic levels (Table 3). Limited data for marine organisms in most cases necessitate the use of freshwater organisms, a limitation also noted by Xu et al. (2013) and Li et al. (2020) in their assessment of pharmaceutical pollutants in coastal environments. We therefore assessed the risk of pharmaceutical compounds, UV filter, and artificial sweeteners utilizing freshwater organisms belonging to three trophic levels (Tables 4, S9, and S10). As shown by Xu et al. (2013) and Biel-Maeso et al. (2018), compounds with an RQ of <0.01 are considered to pose an insignificant risk to aquatic organisms. Those with an RQ higher than 0.01 but lower than 1 pose a low risk, while compounds with an RQ of 1 upward are considered to pose a high risk to aquatic life.

Generally, bezafibrate, caffeine, carbamazepine, diclofenac, lincomycin, and propranolol posed no risk to marine algae (Table 3). The concentrations of fluoxetine in all sampling stations are detrimental to *Skeletonema marinoi*, as observed by the high RQ. Ibuprofen in the Jade Bay and the Weser estuary also poses a high risk to *Skeletonema marinoi*. In addition, concentrations of the antibiotics, clindamycin, clarithromycin, sulfamethoxazole, and trimethoprim measured in all sampling stations posed low risks to *Synechococcus leopoliensis* and *Skeletonema marinoi* with an RQ ranging from 0.02 to 0.45. Furthermore, the concentrations of erythromycin measured in Jade Bay and NS_8 samples pose a low risk to the cyanobacteria *Anabaena* sp. with an RQ of 0.52 and 0.42 in both stations, respectively.

The concentration of caffeine measured at all sampling stations poses high risks to the marine polychaete *Arenicola marina* (Table 3). Caffeine, the most widely detected stimulant in coastal waters, could cause stress and increase the lysosomal content of neutral lipids in hemocytes and the digestive glands of *Mytilus galloprovincialis* (Capolupo et al., 2016). 10 ng L^{-1} of caffeine was also studied to cause oxidative stress and hinder the reproductive functions of the polychaete *Paracentrotus lividus*

Table 3
Risk assessment for pharmaceutical compounds for marine organisms. RQ = Risk Quotient, compounds with an RQ <0.01(in green) pose an insignificant risk to aquatic organisms. Those with an RQ >0.01 but <1 (in yellow) pose a low risk. Compounds with an RQ >1 (in red) pose a high risk. 'na' = not available, 'n.d' = not detected.

Trophic level	Pharmaceutical Compound	Organism	RQ Jade Bay	RQ Weser estuary	RQ NS_7	RQ NS_8
Algae	Atenolol	*Skeletonema marinoi*	na	na	na	na
	Bezafibrate	*Anabaena* sp.	0.00	0.00	0.00	0.00
	Caffeine	Isochrysis galbana	0.00	0.00	0.00	0.00
	Carbamazepine	*Synechococcus leopoliensis*	0.00	0.00	0.00	0.00
	Clarithromycin	*Synechococcus leopoliensis*	0.07	0.03	0.06	0.02
	Clindamycin	*Synechococcus leopoliensis*	0.03	0.11	0.05	0.03
	Diclofenac	*Synechococcus leopoliensis*	0.00	0.00	0.00	0.00
	Erythromycin	*Anabaena sp*	0.52	n.d	n.d	0.42
	Fenofibrate	*Skeletonema marinoi*	na	na	na	na
	fluoxetine	*Skeletonema marinoi*	3.33	2.44	3.89	3.70
	Gemfibrozil	*Skeletonema marinoi*	na	na	na	na
	Ibuprofen	*Skeletonema marinoi*	1.06	1.05	0.32	0.31
	ketoprofen	*Skeletonema marinoi*	0.44	n.d	n.d	0.44
	Lincomycin	*Synechococcus leopoliensis*	0.00	n.d	n.d	0.00
	Metoprolol	*Skeletonema marinoi*	na	na	na	na
	Naproxen	*Skeletonema marinoi*	na	na	na	na
	Propranolol	*Synechococcus leopoliensis*	0.00	n.d	n.d	0.00
	Sulfamethoxazole	*Synechococcus leopoliensis*	0.03	0.04	0.04	0.02
	Trimethoprim	*Skeletonema marinoi*	0.75	0.36	0.41	0.42
	Tylosin	*Synechococcus leopoliensis*	0.03	0.00	0.04	0.02
Invertebrate	Atenolol	*Artemia salina*	n.d	n.d	0.02	0.01
	Bezafibrate	*Artemia salina*	0.00	0.00	0.00	0.00
	Caffeine	*Arenicola marina*	5.44	5.68	2.92	2.92
	Carbamazepine	*Artemia salina*	0.00	0.00	0.00	0.00
	Clarithromycin	*Artemia salina*	0.00	0.00	0.00	0.00
	Clindamycin	*Artemia salina*	0.02	0.05	0.01	0.01
	Diclofenac	*Artemia salina*	0.00	0.00	0.00	0.00
	Erythromycin	*Mytilus edulis*	56.92	n.d	n.d	46.70
	Fenofibrate	*Artemia salina*	0.04	0.02	0.02	0.01
	fluoxetine	*Mytilus californianus*	0.48	0.35	5.29	4.97
	Gemfibrozil	*Artemia salina*	0.11	0.07	0.08	0.37
	Ibuprofen	*M. galloprovincialis*	0.00	0.00	0.00	0.00
	ketoprofen	*Mytilus edulis*	0.00	n.d	n.d	0.00
	Lincomycin	*Navicula ramisissima*	0.00	n.d	n.d	0.00
	Metoprolol	*Artemia salina*	0.03	0.22	0.02	0.01
	Naproxen	*Artemia salina*	1.70	0.02	0.02	1.70
	Propranolol	*Artemia salina*	0.00	n.d	0.00	0.00
	Sulfamethoxazole	*Artemia salina*	0.00	0.00	0.00	0.00
	Trimethoprim	*Mytilus edulis*	0.01	0.01	0.01	0.01

Table 4
Risk assessment for pharmaceutical compounds, UV filters, and artificial sweeteners for freshwater invertebrates. RQ = Risk Quotient, compounds with an RQ <0.01(in green) pose an insignificant risk to aquatic organisms. Those with an RQ >0.01 but <1 (in yellow) pose a low risk. Compounds with an RQ >1 (in red) pose a high risk. 'na' = not available, 'n.d' = not detected.

Micropollutant group	Compound	Invertebrate	RQ Jade Bay	RQ Weser estuary	RQ NS_7	RQ NS_8
Pharmaceutical compound	Bezafibrate	*Hydra attenuata*	0.00	0.00	0.00	0.00
	Caffeine	*Daphnia magna*	0.00	0.00	0.00	0.00
	Carbamazepine	*Thammocephalus platyurus*	0.00	0.00	0.00	0.00
	Chloramphenicol	*Daphnia magna*	0.00	na	0.00	na
	Clarithromycin	*Thammocephalus platyurus*	0.00	0.00	0.00	0.00
	Clindamycin	*Daphnia magna*	0.00	0.00	0.00	0.00
	Cyclamate	*Daphnia magna*	44.17	13.65	n.d	n.d
	Diclofenac	*Daphnia magna*	0.00	0.00	0.00	0.00
	Doxycycline	*Daphnia magna*	0.04	n.d	n.d	0.02
	Erythromycin	*Thammocephalus platyurus*	0.00	n.d	0.00	0.00
	Fenofibrate	*Hydra attenuata*	0.00	0.00	0.00	na
	Fenoprofen	*Daphnia magna*	0.00	n.d	n.d	0.00
	Fluoxetine	*Cladocerans*	0.00	0.00	0.04	0.04
	Gemfibrozil	*Hydra attenuata*	0.00	0.00	0.00	0.00
	Glibenclamide	na	na	na	na	na
	Hydrochlorothiazide	na	na	na	na	na
	Ibuprofen	*Thammocephalus platyurus*	0.00	0.00	0.00	0.00
	Indomethacin	*Thammocephalus platyurus*	0.00	0.00	0.00	0.00
	Ketoprofen	*Thammocephalus platyurus*	0.00	n.d	n.d	n.d
	Lincomycin	*Daphnia magna*	0.00	n.d	n.d	n.d
	Mefenamic Acid	*Thammocephalus platyurus*	0.00	0.00	0.00	0.00
	Metoprolol	*Daphnia magna*	0.00	0.00	0.00	0.00
	Metronidazole	*Daphnia magna*	0.00	na	na	na
	Nadolol	*Daphnia magna*	0.00	na	na	0.00
	Naproxen	*Thammocephalus platyurus*	0.01	0.00	0.00	na
	Novobiocin	*Hydra attenuata*	0.00	0.00	0.00	0.00
	Ofloxacin	*Ceriodaphnia dubia*	0.00	na	0.00	na
	Ornidazole	*Daphnia magna*	0.00	n.d	n.d	0.00
	Oxytetracycline	*Daphnia magna*	0.01	n.d	n.d	n.d
	Salicylic acid	*Daphnia magna*	0.00	0.00	0.00	0.00
	Sulfadiazine	*Daphnia magna*	0.00	0.00	n.d	n.d
	Sulfamethoxazole	*Daphnia magna*	0.00	0.00	n.d	0.00
	Trimethoprim	*Hydra attenuata*	0.00	0.00	0.00	0.00
	Tylosin	*Daphnia magna*	0.00	0.00	0.00	0.00
UV filter	Benzophenone-4	*Isochrysis galbana*	0.33	1.11	0.01	0.06
Artificial sweetener	Acesulfame	*Daphnia magna*	3.99	7.17	1.57	2.22
	Saccharin	*Daphnia magna*	10.48	16.77	2.53	5.23
	Sucralose	*Daphnia magna*	160.12	50.45	99.57	92.85

(Aguirre-Martínez et al., 2015). Indeed, erythromycin concentrations in Jade Bay and NS_8 samples are detrimental to marine mollusk *Mytilus edulis*. The effects of low concentrations of antibiotics on aquatic microbiota have also been a major concern. This is due to the bioactivities and observed toxicities of different antibiotics to marine organisms (González-Pleiter et al., 2013). Marine organisms such as cyanobacteria are primary producers, and as such, they play important roles in marine food webs and nutrient cycling. Cyanobacteria are, however, very sensitive to different concentrations of antibiotics in marine environments (González-Pleiter et al., 2013).

Low antibiotic concentrations of antibiotics also impose selective pressure on the bacterial community and result in the development of resistant bacteria in marine ecosystems (Adenaya et al., 2023). Since low concentrations of antibiotics impose selective pressure on marine bacterial communities and lead to the spread of antibiotic-resistant bacteria in marine environments (Adenaya et al., 2023), it is not surprising that some bacterial strains isolated from Jade Bay showed resistance toward low concentrations of clinically important antibiotics (Adenaya et al. unpublished data). The widespread of low antibiotic concentrations in marine environments also reduces bacterial diversity (van Overbeek et al., 2002), which is important for ecological resilience.

Almost all pharmaceutical compounds detected in samples collected from Jade Bay, the Weser estuary, NS_7, and NS_8 pose insignificant risks to freshwater organisms as the RQ values for each compound were lower than 0.01 (Table 4, S9, and S10). The concentrations of cyclamate measured in the samples collected from Jade Bay and Weser estuary are, however, detrimental to the invertebrate *Daphnia magna* (Table 4). The concentration of the UV filter benzophenone-4 detected in Jade Bay posed low risks to *Isochrysis galbana* and *Paracentrotus lividus*. The samples from the Weser estuary had the highest concentration of benzophenone-4, and the concentration of this compound in this station posed a high risk to the algae *Isochrysis galbana*, with an RQ of 1.11. UV filters are used to protect the skin from UV radiation. Benzophenone-4 and other UV filters, such as benzophenone-3 and methylbenzylidene-camphor, are incorporated into sunscreens like creams and shampoos because they absorb UV radiation (Paredes et al., 2014). Beach bathing is the primary route of UV filters discharge into coastal waters (Fent et al., 2010). He et al. (2019) showed that 0.1–1000 μg L^{-1} of UV filters resulted in the bleaching and mortalities of a marine coral *Seriatopora caliendrum*.

The concentrations of all artificial sweeteners measured in this study posed high risks only to the freshwater invertebrate *Daphnia magna*, as indicated by the high RQ (Table 4). Artificial sweeteners are contributors to global marine pollution (Naik et al., 2021), and even though they are considered to pose little or no health risk to humans at typical concentrations, they pose a prolonged threat to marine ecosystems and are toxic to marine lives (Sang et al., 2014). For instance, 50 ng L^{-1} of sucralose, which is persistent in marine environments, was studied to alter the embryonic development of an aquatic fish and induced malformations that led to their death (Colín-García et al., 2022). The

concentration of all artificial sweeteners measured in this study except sucralose posed no risk to freshwater fish. Sucralose concentrations in Jade Bay, NS_7, and NS_8 posed a low risk to the fish *Danio rerio* (Table S10). Limitations of our study include the recognition that the foam sample was only collected from Jade Bay, which could have contributed to the concentrations of pharmaceutical compounds collected at this station. However, accumulating some pharmaceutical compounds (especially antibiotics and artificial sweeteners) in foam raises concerns about potential environmental impacts and requires further research. This is because waves and currents can transport sea foam and may transport these micropollutants to coastal areas and beaches.

4. Conclusions

The occurrence and distribution of pharmaceutical compounds, UV filters, and artificial sweeteners in coastal waters have continued to receive attention in the past two decades because they are emerging pollutants with societal concerns. The current study is the first to show the spatial and vertical distribution of these contaminants in the near-surface layer and SML in the southern North Sea. Our results show that low concentrations of pharmaceutical compounds span coastal waters of the southern North Sea, especially coastal regions that receive WWTP and freshwater inputs. However, these concentrations were up to those detected in other marine environments such as Cadiz Bay (Biel-Maeso et al., 2018) and Mahdia coastal areas (Afsa et al., 2020). Our study also shows that high concentrations of a UV filter and artificial sweeteners span these coastal environments emphasizing the broader scope of marine environmental pollution.

Even though similar pharmaceutical groups were found in coastal areas and open sea environments, the concentrations of pharmaceutical compounds differed with station and depth. The concentration of pharmaceutical compounds in the SML was higher than in other depths. Also, in three out of four sampling stations, the concentration was low at a depth of 50 cm. Consistent input of low concentrations of pharmaceutical compounds and the hydrophobic nature of the SML may cause their accumulation in the SML, adversely affecting the sustainability and resilience of aquatic lives that reside therein. This can also predispose the environment to ecological risk over a long period. Therefore, it might be worthwhile to consider taking the SML samples and samples from other depths in parallel to obtain a more comprehensive understanding of the distribution of pharmaceutical compounds and their potential impacts on marine ecosystems.

Data on the ecotoxicity of emerging pollutants to marine species are limited and the combined effects of different compounds are unclear. Our study underlines the need for further research to better understand the ecological consequences based on widespread UV filters and artificial sweeteners which were present in almost all our samples. Since we targeted only benzophenone-4, further studies may be required to investigate the distribution of other UV filters in the southern North Sea to assess their ecotoxicological effects on marine life. Also, since antibiotics and artificial sweeteners detected in our study pose low and high risks to marine aquatic algae and invertebrates, more effort is required to curb the effects of pharmaceutical compounds and micropollutants of emerging concern on the biodiversity of the southern North Sea.

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CRediT authorship contribution statement

Adenike Adenaya: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Ruben Rios Quintero:** Methodology. **Thorsten Brinkhoff:** Writing – review & editing, Conceptualization. **Pablo A. Lara-Martín:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Oliver Wurl:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Mariana Ribas-Ribas:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The online version of our data can be found here: https://doi.org/10.1594/PANGAEA.963943.

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Appendix A. Supplementary data

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